

OPTICAL GROUND-TRUTH LANDING MODULE

Phase 5 - Physical Commissioning: Field Datasheet & Checklist

Optical.md v1.5 - Politecnico di Milano - Measurements for Mechanical Engineering (A.Y. 2025-26)

Updated for WO-OPT-4 (tennis→basketball substitution, closed 2026-07-10): items marked **[WO-OPT-4]** below are new or amended by that work order and are carried forward from its closeout into this sheet.

Purpose: recover the landing point from one fixed phone high-speed camera (video in, (r, theta) out) and cross-validate it against the three-tape multilateration reference. This sheet is both a NOTE (write every value) and a CHECKLIST (tick every box). It is the interface between operators (section 8). Work top to bottom: [C0] then C1..C7, then the mode decision, then the campaign log. Fill owners/initials as you go.

Session identification

Session date: _____ Venue / location: _____

Operators (initials): _____ Reviewer (array conventions): _____

Sky / lighting: _____ Wind (calm required; note): _____

Start time: _____ End time: _____ Ambient temp [C]: _____

Main-pipeline session IDs this session maps to: _____

☐ Manual folder backup of the WHOLE project taken BEFORE this session (O1; no git).

0. Rules & discipline (read before starting)

- ☐ Use the venv interpreter for EVERY command: `.\venv\Scripts\python.exe` (never bare python / pytest).
- ☐ `raw.mp4` is SACRED: verbatim from the phone, never modified or overwritten. Every other artefact is derived and regenerable.
- ☐ `--fps-measured` is MANDATORY on every real-footage `process_clip.py` run (pass the C1 stopwatch-verified rate). The nominal rate is never used for timing (section 3.2 / 8).
- ☐ The camera must NOT move between calibration and the last throw of a session (section 3.1c). If bumped, re-calibrate and re-run the stability check.
- ☐ Throws are NOT steered toward the camera (O10). If natural in-FOV coverage < 8, APPEND extra camera-sector throws to the campaign; never substitute.
- ☐ The section-8 GO/NO-GO gate ($\text{RMS} \leq 1.5 \text{ cm}$ over ≥ 8 points) is PRE-REGISTERED: fixed before data and NOT renegotiated after seeing results.
- ☐ Any clip that contains a bounce keeps ≥ 3 frames AFTER the bounce (image-v) peak when trimming (section 9 / D14) - otherwise the reversal cannot be confirmed and a silent late-timing bias returns.
- ☐ **[WO-OPT-4]** Basketball contact duration is UNVERIFIED (D21) - it may span MORE frames at 480 fps than the tennis ball did (never assumed zero/one frame to begin with, but the margin this needs is now unknown). Treat the ≥ 3 -frame rule above as a FLOOR, not a target: if the bounce footage shows the ball flattened/in contact for more than a couple of frames, trim with MORE post-peak margin, and do not quote any millisecond contact-duration figure in the report until it is measured from this session's own footage or a primary source.
! Outdoors, daylight only. Prefer high overcast. Avoid direct low sun (hard moving shadows and HSV washout). Verify motion blur visually in the first test clip of each mode.

[C0] Phone re-verification gate (do this FIRST)

The whole pipeline was built against the OnePlus 12R. Re-confirm the device actually being used before any physical testing. Switch only if a better phone WINS on criterion 1 and improves the rest. The pipeline is

device-agnostic.

Device in use (make / model): _____

- ☐ Criterion 1 (NON-NEGOTIABLE): slow-mo frames are REAL captured frames, not interpolated. Verified by inspection / spec.

Highest REAL frame rate at a usable resolution: _____ fps @ _____

Exposure / shutter lock available in slow-mo? [] yes [] no -> if yes, LOCK it.

Slow-mo crop (least aggressive preferred) - note: _____

Decision: [] keep OnePlus 12R [] switch to: _____

! If SWITCHED: redo intrinsics (section 3.4 / C2) and frame-rate verification (section 3.2 / C1) for the new device. Nothing else changes.

Owner: _____ Date / initials: _____

[C1] Frame-rate verification - BOTH modes (section 3.2)

Datasheet fps values are NOT ground truth. Film a millisecond stopwatch (phone/laptop screen) in each slow-mo mode and step through frames. Also read the container-delivered rate; the MEASURED value feeds all timing math.

Mode 720p @ 480 fps

- ☐ PASS: distinct, monotonically advancing timestamps every frame; no ghosted / blended digits; no repeated values.

Container-reported (delivered) fps: _____ Nominal: 480

C1 stopwatch-verified rate (use as --fps-measured): _____ fps

Mode 1080p @ 240 fps

- ☐ PASS (same criteria as above).

Container-reported fps: _____ Nominal: 240 C1 verified: _____ fps

- ☐ Container vs measured vs nominal agree within capture.fps_tol_pct (0.5%)? (beyond -> fps_mismatch flag).
- ☐ Checked for GROSS disagreement > capture.fps_refuse_pct (20%): a playback-vs-capture rate mix-up makes the pipeline REFUSE and write nothing (section 3.2 / D9).

! A FAIL in a mode BANS that mode from the project.

Modes passed: [] 720p480 [] 1080p240 Owner: _____

[C2] Checkerboard intrinsics - per used mode (section 3.4)

cv2.calibrateCamera, once per camera per video mode actually used (the slow-mo crop differs between modes, so intrinsics differ). Print a checkerboard; measure the square size with a rule. Capture 15-20 stills at varied poses filling the frame.

Checkerboard inner corners (e.g. 9x6): _____ Measured square size [m]: _____

- ☐ 15-20 stills captured at varied poses filling the frame, for each used mode.

720p480: RMS reproj error [px]: _____ calib/intrinsics_720p480.yaml written []

1080p240: RMS reproj error [px]: _____ calib/intrinsics_1080p240.yaml written []

- ☐ All measurement frames will be undistorted (cv2.undistort) before any homography use.

Owner: _____

[C3] Tripod geometry & camera position (section 3.1)

Start ~1.2-1.5 m elevation at ~3-4 m horizontal standoff from the centroid. Constraints in PRIORITY order: (a) all markers AND the expected landing sector fit inside the cropped slow-mo FOV; (b) the floor is viewed obliquely enough that the homography stays well-conditioned (grazing views compress the far field and blow up per-pixel uncertainty); (c) the camera never moves from calibration to the last throw.

☐ All homography markers AND the expected landing sector fit inside the CROPPED slow-mo FOV (verified on-site, not from the normal camera preview).

☐ View is oblique but NOT grazing.

Camera height above floor [m] (tape): _____ Horizontal standoff [m]: _____

Bearing / orientation note: _____

Tape-measured camera centre in ARRAY frame: X = _____ Y = _____ Z = _____ [m]

! These (X, Y, Z) may be passed to `calibrate_homography.py --camera-measured X Y Z` for the decomposed-vs-measured cross-check (informational only; section 5.4 / D13).

☐ Exposure / shutter locked if the mode allows; blur checked in the first test clip.

Owner: _____

[C4] Marker survey (section 3.3)

ArUco markers (DICT_4X4_50), ~6-8, laid flat and spread near/far AND left/right across the used image region. Plus surveyed pod floor marks + the centroid mark. Survey every position in ARRAY coordinates (tape from the centroid and the S1/S2/S3 floor marks). Designate at least ONE held-out check point.

☐ Markers spread near/far AND left/right (conditions the fit; absorbs residual distortion).

☐ At least ONE surveyed marker designated HELD-OUT (excluded from the fit; integrity check).

☐ Held-out check point is an ArUco marker (so the section-5.1 drift re-check can auto-detect it; a manual mark reports `stability_unchecked`).

☐ **[WO-OPT-4]** A resting basketball (radius ~0.12 m, i.e. ~24 cm across) can OCCLUDE a floor fiducial - confirmed against the module's own static-validation geometry, where it hid the held-out check point outright. When laying out and surveying markers, keep EVERY marker - especially the held-out check point - clear of anywhere the ball will be placed for C6/C7 or is likely to come to rest after a bounce.

Held-out check-point id: _____

Measured survey sigma [m] (expect +/- 2-3 mm): _____

Marker survey table (populate `config.yaml` `markers.fit_points` with id / xz / held_out before calibrating):

id	x [m]	z [m]	held_out?	detected?	notes
-----					-----
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

Owner (reviewed by someone who knows the array conventions): _____

[C5] Homography calibration + stability (section 5.1)

Run `scripts/calibrate_homography.py`: `undistort -> auto-detect ArUco -> click the pod/centroid reference marks -> plain-LS fit (held-out excluded) -> write calib/homography_<id>.yaml. Optionally pass --intrinsic and --camera-measured (C3).`

calib_id: _____ Mode: _____

Reprojection RMS [px]: _____

Held-out check-point error [mm] (tol check_point_tol_m = 5 mm): _____ OK? []

! If the check-point guard trips (check_point_fail): the survey or the frame is suspect - re-measure; do NOT trust that calibration.

☐ Marker conditioning OK (no cond_warn; markers well spread).

Stability re-check (section 5.1 step 5: drift AND absolute error)

Check-point mapping error vs surveyed: START = _____ mm END = _____ mm (tol 5)

Start-to-END drift [mm] (tol 5 mm): _____

☐ No homography_drift (start-vs-end drift <= 5 mm): the camera did not move during the session.

☐ No stability_abs_fail (absolute error <= 5 mm at BOTH frames): the calibration is not stale / was not bumped between calibration and the clip.

(optional) camera_delta_m norm from --camera-measured [mm]: _____ (~12 mm order)

Owner: _____

[C6] Detector characterisation (section 7 item 3)

Place the ball statically at >= 3 surveyed positions and measure the pixel-localisation scatter (feeds the uncertainty budget). Tune the HSV band against REAL footage WITH shadows, not only clean clips.

Pos 1 xz [m]: _____ detected/frames: _____ sigma_px: _____

Pos 2 xz [m]: _____ detected/frames: _____ sigma_px: _____

Pos 3 xz [m]: _____ detected/frames: _____ sigma_px: _____

Adopted pixel-localisation sigma [px]: _____

☐ **[WO-OPT-4]** BOTH the HSV band AND the detector area band shipped in config.yaml are UNVERIFIED placeholders for the basketball (D18/D19), not just HSV as in earlier commissioning rounds - the area band was re-derived from SYNTHETIC measurements only. Tune and record both against this session's real footage, not HSV alone.

☐ HSV band tuned against real footage WITH shadows.

Tuned HSV lower [H,S,V]: _____ upper [H,S,V]: _____

☐ Area band re-tuned against real footage (detect.min_area_px / max_area_px - the shipped 30 / 245000 is a synthetic-derived placeholder, D19).

Tuned area band [px^2] lower: _____ upper: _____

☐ **[WO-OPT-4]** Ball-compression sigma (uncertainty.sigma_h_m, the parallax height surrogate at contact) is a basketball quantity that is UNMEASURED (D20) - and unlike the other four budget placeholders below, the spec does not define HOW to measure it. There is no prescribed procedure for this session. Decide and record a method here (e.g. high-speed footage of the ball's deformation depth at contact, or a documented literature/manufacturer figure) before flipping uncertainty.sigma_h_m.measured to true - do not leave the config value unchanged and flip the flag regardless.

Owner: _____

[C7] Static-point validation - THE GATE (section 8, pre-registered O9)

Place the ball statically at >= 8 INDEPENDENT surveyed points (NOT the ultrasonic characterisation set), spread across the working FOV including near/far and off-axis. Run each through the full pipeline (validate_static.py).

Compare against the three-tape multilateration solve at the same points. Run in BOTH modes.

GATE: GO if RMS disagreement <= 1.5 cm over the >= 8 points. Otherwise NO-GO. (config tolerances.rms_gate_m = 0.015)

[**WO-OPT-4**] As with C4: verify none of these 8+ static points, nor the ball resting between shots, occludes a fit or held-out marker (WO-OPT-4 finding, see [C4] above).

[illegible]

- ☐ Session stability re-check across the WHOLE gate span is clean (no drift, no abs_fail).
- ! NO-GO in BOTH modes = the module is reported as a characterised NEGATIVE result with its diagnosis (chalk-mark diagnostics, section 5.4). No campaign integration is attempted. Do NOT renegotiate the gate.
- Owner: _____

Mode decision & chalk cross-check (sections 4, 5.4)

720p/480 is the working default (it halves the temporal quantum). Keep whichever mode yields the LOWER static-point RMS; record BOTH in the commissioning report.

Chosen campaign mode: ☐ 720p480 ☒ 1080p240 Reason (RMS values): _____

Chalk-mark cross-check (commissioning-only): during C7 and the first few live throws, ALSO map the chalk mark through H (a zero-parallax floor observable). chalk-vs-tape isolates homography error; tracker-vs-chalk isolates tracking/parallax error. Retired after commissioning.

- ☐ Chalk cross-check performed on C7 points and the first live throws.

Campaign throw log (sections 8 / O10, 9)

Commit > 8 in-FOV ground-truthed throws. Session IDs match the main pipeline (optical_<YYYY-MM-DD>_T##). Record trim in/out frames (keep >= 3 frames after any bounce peak - more if the contact looks drawn out, see the [WO-OPT-4] note in section 0), whether the landing is in-FOV, and wind/lighting per throw.

[illegible]

Config values to finalise (flip measured:true after commissioning)

Until each is flipped, every throw is (correctly) flagged unmeasured_uncertainty_components. Edit config.yaml, then re-run.

- ☐ uncertainty.pixel_sigma_px.value <- C6 localisation scatter; set measured: true
- ☐ uncertainty.sigma_h_m.value <- ball compression at contact; set measured: true. **[WO-OPT-4]** this is now a BASKETBALL figure and the spec defines no measurement procedure for it (D20) - see the [C6] note above; do not flip this flag without recording how the value was obtained.
- ☐ uncertainty.sigma_C_m.value <- repeated C5 calibrations + tape; set measured: true
- ☐ uncertainty.rolling_shutter_m.value <- falling-ball / plumb-line film; set measured: true
- ☐ markers.survey_sigma_m.value <- C4 tape survey sigma; set measured: true
- ☐ capture.fps_tol_pct tightened once C1 characterises the device's rate stability.
Rolling-shutter characterisation value [m]: _____ (film a falling ball beside a plumb line / stopwatch; estimate skew - do NOT assume it away)

Data management & transfer (section 9)

- ☐ Slate every clip: the first ~1 s films a paper slate with the session/throw ID handwritten (footage is self-identifying).
- ☐ Transfer via USB-C plain MTP copy to the PC immediately after the session. NO cloud round-trips (some services re-encode video).
- ☐ After copy, VERIFY the imported file's frame-rate metadata matches capture (fps_measured) before trusting timing (a transfer re-encode corrupts the temporal base).
- ☐ Files renamed on copy into data/optical/<session_id>/raw.mp4 ; raw.mp4 untouched thereafter.
- ☐ Manifest entry written per clip: process_clip.py --notes "wind/lighting; operator initials" (wind/lighting/initials go in the free-text notes field).

Cautions / risk register (carry into the manifest notes)

! Wind: calm conditions required; a steady crosswind violates the parabola assumption. Note the wind per session.

! Ground planarity: choose smooth, level paving; deviation shows up as position-dependent error in the C7 static validation.

! Sun angle / shadows: avoid direct low sun; hard moving shadows and HSV washout degrade detection. Verify blur in the first clip.

! Bounce clips: keep ≥ 3 frames after the image-v peak, or the reversal cannot be confirmed and the +0.5-frame late bias silently returns.

! Grazing camera view compresses the far field and blows up per-pixel uncertainty - keep the view oblique, not grazing.

! Calibration click-UI: no undo; ZOOM before clicking; a mis-click means restarting the clicking pass. Do a dry run before the field session.

[WO-OPT-4] Occlusion (new, WO-OPT-4): a resting basketball (~24 cm across) can hide a floor marker. Keep the held-out check point and all fit markers clear of anywhere the ball is placed for C6/C7 or comes to rest after a throw.

[WO-OPT-4] Contact duration (new, WO-OPT-4): unlike the tennis ball, basketball contact duration at 480 fps is unverified and may span multiple frames (D21) - do not assume it away when trimming or quoting timing figures; measure it from this session's footage if possible.

Notes for the Phase 5 report (informational - not required this session)

[WO-OPT-4] *The apparent-size (pixel-radius) height cue for sub-frame contact timing was rejected for the tennis ball as signal-starved (its pixel radius barely changed across the descent window). The basketball's much larger apparent size and per-frame radius change MAY make this cue viable. Not required or implemented for this commissioning session - if useful, a data-processing pass over this session's real footage can re-evaluate it after the fact (see IMPLEMENTATION_NOTES_OPTICAL.md, WO-OPT-4 Stage 5).*

Sign-off

- ☐ Commissioning report produced (both modes: frame-rate verification, intrinsics RMS, homography RMS + check point, detector scatter, static-point RMS vs gate, mode decision).

GO / NO-GO decision (human; recorded in IMPLEMENTATION_NOTES_OPTICAL.md): _____

Signed (operator): _____ Date: _____

Signed (reviewer): _____ Date: _____